

Notes: Hart and Holmström (QJE, 2010)

A Theory of Firm Scope

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1 Big picture

This note summarizes Hart and Holmström's theory of firm scope. The paper asks a classic question in the theory of the firm: when should two activities remain in separate firms, and when should they be brought under one authority? The paper departs from the standard property-rights model by emphasizing *authority*, *private benefits*, and *aggrievement/shading* rather than only ex post bargaining over asset ownership.

The central economic idea is simple. Two units may need to coordinate on a common standard or platform. If they remain separate, each unit manager has decision rights over his own unit and may veto coordination if the distribution of gains is unfavorable. This creates too little coordination. If the two units are integrated under one boss, the boss internalizes the profit externality across units and can force coordination. But the boss puts too little weight on the managers' private benefits, so integration creates too much coordination and too much aggrievement.

1.1 One-sentence summary

This paper develops a reference-point theory of firm scope in which nonintegration preserves local private benefits but fails to internalize coordination externalities, whereas integration centralizes authority and internalizes profits but underweights employee private benefits, implying a trade-off between too little coordination in markets and too much coordination inside firms.

1.2 Why this paper matters

The paper is important because it provides a theory of lateral firm boundaries. Much of the property-rights literature focuses on vertical integration and relationship-specific investments. Hart and Holmström instead focus on strategic coordination across related activities, such as whether two technologies should adopt a common standard. This makes the theory especially useful for understanding platform leadership, technology integration, acquisitions, and internal organizational design.

2 Research question

The paper asks three related questions:

1. Why can integration improve coordination between two activities?
2. Why can integration also be costly even when it internalizes externalities?
3. How can delegation inside an integrated firm provide an intermediate governance structure between market contracting and centralized authority?

The authors' answer is that firm boundaries determine who has authority over noncontractible strategic decisions. Authority matters because parties may simply exercise their decision rights rather than bargain ex post. Organizational form therefore affects actual decisions, not merely investment incentives.

3 Incremental contribution of the paper

3.1 Contribution relative to the standard property-rights theory

The standard Grossman-Hart-Moore property-rights framework studies how ownership affects ex ante relationship-specific investment incentives through ex post bargaining positions. In that framework, ex post efficient bargaining is usually assumed, so ownership affects incentives but does not directly determine the ex post decision.

Hart and Holmström make four incremental contributions:

1. They remove the assumption that ex post conflicts are solved by efficient bargaining with side payments. Instead, parties often exercise authority directly.
2. They embed firm boundaries in the “contracts as reference points” framework. Organizational form affects parties’ entitlements, and disappointed parties shade.
3. They introduce private benefits as a central force in firm scope. Employees and managers care about independence, technology, career paths, identity, and job satisfaction.
4. They develop a theory of delegation as credible but imperfect authority transfer inside the firm. Delegation matters because reversal is costly: it creates aggrievement.

3.2 Contribution to organizational economics

The paper moves the analysis from ownership of assets toward control of activities. It captures a practical organizational issue: firms are often defined not by legal asset ownership alone, but by what they do, which technologies they combine, and whose authority governs the strategic direction.

3.3 Contribution to platform and technology strategy

The Cisco-StrataCom application shows how acquisitions can be used to control platform evolution. The issue is not only avoiding hold-up. It is also about deciding the path of technological integration, managing employee private benefits, and reducing conflict around a common standard.

4 Core intuition

Consider two units, A and B. Each can choose Yes or No. Coordination occurs only if both choose Yes. The decision may represent whether both units adopt the same platform, standard, or technology interface.

- Under nonintegration, each unit manager controls his own decision. If one unit loses from coordination, that unit can veto it. This produces too little coordination.
- Under integration, a single boss controls both decisions. The boss internalizes total profit across units and can force coordination. But the boss ignores or underweights unit-level private benefits. This produces too much coordination.
- Under delegation, the boss gives decision rights to unit managers, but can reverse delegation if the stakes are high enough. Delegation can reduce overcoordination, but reversal creates extra aggrievement.

The theory therefore gives a sharp and memorable trade-off:

Nonintegration is too parochial; integration is too insensitive to private benefits.

5 Model setup

5.1 Players and timing

There are two units, A and B. Each unit has a manager. Each unit chooses either Yes (Y) or No (N). Coordination occurs only when both choose Yes.

The timing is:

1. An organizational form is chosen: nonintegration, integration, or delegation.
2. The relevant decision makers choose Y or N.
3. Payoffs are realized, including possible shading losses.

5.2 Payoffs

Each unit generates two types of benefit:

1. Monetary profit, denoted v .
2. Private benefit, denoted w .

Private benefits represent job satisfaction, independence, technological identity, career prospects, and other nontransferable benefits enjoyed by managers or workers.

The change in total surplus for unit A from coordination is

$$\Delta z_A \equiv \Delta v_A + \Delta w_A,$$

and similarly for unit B,

$$\Delta z_B \equiv \Delta v_B + \Delta w_B.$$

The first-best rule, ignoring shading costs, is:

$$\Delta z_A + \Delta z_B > 0.$$

Thus coordination is efficient if the aggregate gain in profits and private benefits is positive.

5.3 Private-benefit assumption

The paper assumes that coordination tends to reduce independence:

$$\Delta w_A \leq 0, \quad \Delta w_B \leq 0.$$

This means that a common standard or platform may be profitable but privately costly for unit managers. For example, workers may have to abandon familiar technology, lose autonomy, or follow another unit's platform.

6 Reference points, aggrievement, and shading

The paper uses the contracts-as-reference-points logic. Organizational form defines what parties feel entitled to. If a party does not get his preferred outcome within the organizational form, he becomes aggrieved.

If a party receives k less than his best feasible payoff, he shades by reducing the other side's payoff by

$$\theta k,$$

where

$$0 < \theta < 1.$$

Shading is socially wasteful. It hurts the shaded-against party but does not benefit the party who shades. Hence it creates a deadweight loss.

This is crucial because it explains why ex post bargaining is costly. Renegotiation reopens entitlements and can generate aggrievement. Therefore the paper rules out costless ex post bargaining.

7 Organizational forms

7.1 Nonintegration without cooperation

Under nonintegration without cooperation, each manager controls his own unit. If the units fail to coordinate, the relationship ends, so shading under noncoordination is not possible.

There are three cases:

1. If both managers dislike coordination, no coordination occurs and this is efficient.
2. If both managers like coordination, coordination occurs and this is efficient.
3. If one manager likes coordination and the other dislikes it, the losing manager vetoes coordination.

The problem is Case 3. Coordination may increase total surplus, but if the gains are unevenly distributed, the loser blocks it.

7.2 Main implication

Nonintegration without cooperation has no shading cost in equilibrium, but it may fail to coordinate when coordination is first-best efficient. Thus it errs on the side of too little coordination.

7.3 Nonintegration with cooperation

Under nonintegration with cooperation, the relationship continues even if coordination fails. This means that the party who wanted coordination can shade against the vetoing party.

Suppose manager i loses from coordination while manager j gains:

$$\Delta z_i < 0, \quad \Delta z_j > 0.$$

Manager i may reluctantly coordinate if the cost of being shaded by manager j is sufficiently large:

$$\Delta z_i + \theta \Delta z_j > 0.$$

If this condition holds, coordination occurs but manager i is aggrieved and shades. If it does not hold, noncoordination occurs but manager j shades. Either way, conflict creates deadweight loss.

7.4 Integration

Under integration, one boss controls both units. The boss cares about total monetary profits:

$$\Delta v_A + \Delta v_B.$$

But the boss does not directly care about private benefits:

$$\Delta w_A + \Delta w_B.$$

If coordination raises total profits, the boss wants to coordinate. But if coordination reduces private benefits, managers are aggrieved and shade. The boss coordinates if

$$\Delta v_A + \Delta v_B + \theta(\Delta w_A + \Delta w_B) \geq 0.$$

Because $\theta < 1$, the boss underweights private-benefit losses relative to the social planner. The first-best condition is

$$\Delta v_A + \Delta v_B + \Delta w_A + \Delta w_B > 0,$$

whereas the integration condition discounts private-benefit losses by θ .

7.5 Main implication

Integration internalizes the profit externality across units, but it underweights private benefits. Therefore integration errs on the side of too much coordination.

8 Main propositions and economic interpretation

8.1 Proposition 1: nonintegration versus integration

The paper's main result is:

Nonintegration errs on the side of too little coordination, whereas integration errs on the side of too much coordination.

More precisely:

- If coordination occurs under nonintegration, it is first-best efficient. But nonintegration may fail to coordinate even when coordination is efficient.
- If coordination is first-best efficient, integration leads to coordination. But integration may coordinate even when coordination is not efficient.
- Nonintegration without cooperation is best when noncoordination is first-best efficient.
- Integration is more attractive when the gains from coordination are large and unevenly distributed.

8.2 Economic intuition

Nonintegration gives each unit strong protection over its private benefits, but this protection creates veto power. Integration removes veto power, but this makes it easier for the boss to impose coordination even when managers suffer private losses.

This is the paper's fundamental trade-off:

Organizational form	Benefit	Cost
Nonintegration	Protects local private benefits; avoids forced coordination	Fails to internalize cross-unit externalities; one unit may veto efficient coordination
Integration	Internalizes profit externalities and enables common direction	Underweights private benefits and can force excessive coordination
Delegation	Creates intermediate authority and partial commitment	Reversal is possible but costly; may generate extra aggrievement

9 Delegation

9.1 Why delegation is hard to model

In many models, delegation is difficult to make credible because the boss can simply take back decision rights ex post. Hart and Holmström solve this by using aggrievement. If the boss delegates and later reverses delegation, subordinates view this as a breach of promise. Reversal raises the shading parameter from θ to $\bar{\theta}$, where

$$1 > \bar{\theta} > \theta.$$

This means delegation is not the same as ownership, but it is not meaningless either.

9.2 Delegation as an intermediate form

Under delegation, unit managers make decisions unless the boss reverses. This can reduce excessive coordination relative to integration because the boss will reverse only when the profit gain from coordination is sufficiently large.

Thus delegation lies between nonintegration and integration:

- It gives managers more local control than integration.
- It gives the boss more ultimate control than nonintegration.
- It is valuable under uncertainty because it allows the boss to intervene only in important states.

9.3 Propositions 2-4

The delegation section establishes three key ideas:

1. Delegation can produce noncoordination, reluctant coordination, or forced coordination.
2. Delegation creates weakly less coordination than integration, but still too much coordination relative to the first-best.
3. Under perfect certainty, delegation is never strictly optimal; under uncertainty, delegation can be strictly valuable.

9.4 Intuition for why uncertainty matters

When the environment is uncertain, the boss may not know *ex ante* whether coordination will be important. Delegation gives the boss a way to commit not to intervene in moderate states where intervention would be inefficient, while preserving the option to reverse delegation in extreme states where coordination is very valuable.

10 Application: Cisco and StrataCom

10.1 The empirical setting

The paper applies the theory to Cisco's 1996 acquisition of StrataCom. Cisco's Internet Operating System (IOS) became a dominant platform in the network industry. StrataCom was a leading provider of Asynchronous Transmission Mode (ATM), a new transmission technology that could potentially threaten Cisco's router-based technology.

Cisco faced a strategic decision: should it make IOS and ATM compatible, and if so, should this be done through market cooperation or acquisition?

10.2 Three options in the model's language

Cisco had three broad options:

1. **Nonintegration without coordination.** Cisco could refuse compatibility and hope ATM would not become important.
2. **Nonintegration with coordination.** Cisco could cooperate with StrataCom and other ATM firms to make the technologies compatible.
3. **Integration with coordination.** Cisco could acquire StrataCom and make IOS and ATM compatible internally.

Cisco chose the third option.

10.3 Why the acquisition fits the model

The case fits the model because the gains from coordination were large but unevenly distributed. Coordination between IOS and ATM would create significant joint value, but StrataCom may have feared that cooperation would tilt the platform game in Cisco's favor. Market coordination therefore faced the classic nonintegration problem: one party may veto efficient coordination because the gains are uneven.

Integration solved this problem by giving Cisco decision rights over both technologies. It could determine the path of IOS-ATM integration internally.

10.4 Private benefits in the Cisco case

The paper also emphasizes that Cisco cared about private benefits. Cisco's employees were biased toward IP technology and resisted ATM. Cisco's sales force disliked ATM partly because it was cheaper and less sophisticated, which reduced commissions. On StrataCom's side, private benefits may have been positive because acquisition gave the technology access to Cisco's customer base.

Cisco's acquisition strategy also recognized private benefits. Chambers emphasized common vision, cultural compatibility, quick shareholder wins, long-term wins for all constituencies, and

geographic proximity. Cisco tried to preserve entrepreneurship inside acquired firms. This is consistent with the model's idea that integration can create private-benefit losses and shading unless authority is managed carefully.

11 Relation to the literature

11.1 Grossman-Hart-Moore property-rights theory

The paper builds on the incomplete-contracts tradition but changes the mechanism. In Grossman-Hart-Moore, ownership improves bargaining position and investment incentives. In Hart and Holmström, authority directly affects ex post decisions and the scope of coordination.

11.2 Hart and Moore reference-point theory

The paper applies the contracts-as-reference-points approach to organizational form. Firm boundaries are not just legal arrangements; they shape entitlements. When actual decisions violate perceived entitlements, shading creates deadweight losses.

11.3 Internal capital markets

The model resembles internal capital-market theories because a central boss can allocate resources across units and internalize total profits. But Hart and Holmström emphasize a different cost of centralization: the boss underweights local private benefits.

11.4 Organizational culture and vision

The paper connects naturally to theories of firm culture, managerial vision, and narrow business strategies. A narrow firm may better align with employees' private benefits. A broad firm can coordinate more effectively but may generate conflict because employees' preferences are more heterogeneous.

12 Extracted figures and tables from the paper

12.1 Figure I: Timeline of organizational form, decision making, and payoff realization; Table I: Coordination payoff matrix

Read. Figure I shows the three-stage timing: organizational form is chosen first, decisions are made second, and payoffs are realized last. Table I formalizes the binary coordination game. Coordination occurs only when both units choose Yes. If either unit chooses No, payoffs are normalized to zero.

Economic message. The timeline clarifies why organizational form matters ex ante: it allocates authority before the relevant strategic decision is taken. The payoff matrix clarifies that coordination is a joint decision with veto power under nonintegration. This simple structure generates the core trade-off between market vetoes and firm authority.

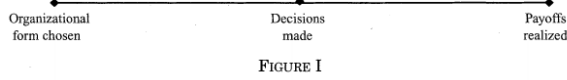


FIGURE I

(a) Figure I: Timeline of organizational form, decision making, and payoff realization

TABLE I
PAYOFFS

	Unit B	
	Y	N
Y	A : $\Delta v_A, \Delta w_A$	A : 0, 0
Unit A	B : $\Delta v_B, \Delta w_B$	B : 0, 0
N	A : 0, 0	A : 0, 0
	B : 0, 0	B : 0, 0

(b) Table I: Coordination payoff matrix

Extracted directly from Hart and Holmström (2010).

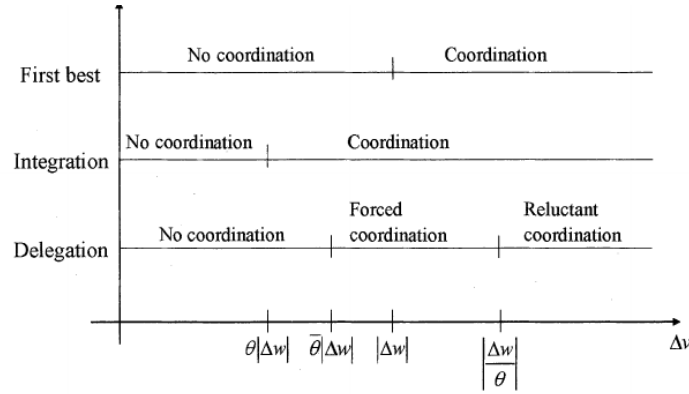


FIGURE II

Figure II: Delegation, integration, and the first-best coordination regions

12.2 Figure II: Delegation, integration, and the first-best coordination regions

Read. Figure II compares the coordination regions under first-best, integration, and delegation. The horizontal axis represents the profit gain from coordination, while the thresholds depend on private-benefit losses and the shading parameters.

Economic message. Integration coordinates sooner than the first-best because the boss underweights private-benefit losses. Delegation delays coordination relative to integration because reversal is costly. Thus delegation can move the organization closer to the first-best when uncertainty makes it valuable to preserve the option to intervene only in high-stakes states.

13 Detailed model walkthrough

13.1 First-best benchmark

The first-best ignores authority distortions and shading. Coordination should occur when

$$\Delta z_A + \Delta z_B = \Delta v_A + \Delta v_B + \Delta w_A + \Delta w_B > 0.$$

This benchmark treats profit changes and private-benefit changes symmetrically.

13.2 Why nonintegration can fail

Under nonintegration, each manager evaluates coordination from his own unit's perspective. If manager A loses and manager B gains, manager A can veto. This may happen even when

$$\Delta z_A + \Delta z_B > 0.$$

Thus nonintegration fails when gains from coordination are large in aggregate but unevenly distributed.

13.3 Why integration can overcoordinate

Under integration, the boss evaluates coordination using total profit plus only a fraction of private-benefit losses:

$$\Delta v_A + \Delta v_B + \theta(\Delta w_A + \Delta w_B) \geq 0.$$

Because $\Delta w_A + \Delta w_B < 0$ and $\theta < 1$, this condition is easier to satisfy than the first-best condition. Therefore integration may coordinate even when total surplus is negative.

13.4 Why delegation can improve on integration

Delegation can be valuable because it increases the boss's cost of intervention. When the boss reverses delegation, the shading parameter becomes $\bar{\theta}$ rather than θ . This makes reversal costly and therefore credible. Delegation is useful when the boss wants to commit not to intervene in moderate states but still wants the option to intervene in extreme states.

14 What to remember

1. Firm boundaries allocate authority over noncontractible strategic decisions.
2. Authority matters because parties often exercise decision rights rather than bargain ex post.
3. Nonintegration protects private benefits but can fail to coordinate because each unit has veto power.
4. Integration internalizes profit externalities but can impose excessive coordination because the boss underweights private benefits.
5. Delegation is meaningful because reversal creates aggrievement; it is not equivalent to full ownership.
6. The Cisco-StrataCom case illustrates how acquisition can solve platform coordination problems while creating private-benefit and cultural-integration concerns.

15 Strengths, limitations, and implications

15.1 Strengths

The paper provides a clean theoretical explanation for why firms acquire related activities even when market coordination is possible. It also explains why integration is not always desirable: centralized authority can ignore local knowledge, identity, and private benefits.

15.2 Limitations

The model is stylized. It has two units, binary decisions, and a reduced-form shading parameter. The treatment of private benefits is powerful but abstract. The Cisco case is illustrative rather than an empirical test.

15.3 Implications

The theory implies that optimal firm scope depends on the distribution of coordination gains, the size of private-benefit losses, the severity of shading, and the credibility of delegation. It is especially useful for studying platform firms, technology mergers, standard-setting, internal restructuring, and organization design.

16 Conclusion

Hart and Holmström develop a theory of firm scope in which the central issue is not only asset ownership or investment incentives, but also authority over strategic coordination. Nonintegration gives parties veto rights and can block efficient coordination. Integration removes vetoes and improves coordination, but it can ignore private benefits and generate aggrievement. Delegation provides an intermediate structure when reversal is costly and uncertainty makes flexibility valuable.

The paper's deeper lesson is that firm boundaries are about the distribution of activities under authority. A firm is not merely a bundle of assets; it is a governance structure that decides which activities are coordinated under a common boss and which remain independent.